

FINAL INCIDENT INVESTIGATION & FORENSIC REPORT

Phishing Management Structure & Proactive AI Perimeter Defense

1. DOCUMENT CONTROL & METADATA

Metadata Field	Incident Case Details
Incident ID	INC-DFIR-2026-0618 (Cross-Ref: PHI-2026-0417)
Target Host System	Windows 10 Workstation Asset (192.168.92.133)
Attacker / C2 Server	Kali Linux / GoPhish Framework Infrastructure (192.168.56.101)
Target Indicator URL	http://192.168.56.101:8080?rid=3D17FCWja
Lead Investigator	Menna Said Ahmed Abdelkader Sabek
Framework Mapping	SANS 6-Phase Incident Handling Model & NIST SP 800-61r3
Classification Level	High — Confirmed Credential Harvesting

2. EXECUTIVE SUMMARY

This comprehensive forensics and response report marks the formal closure of incident **INC-DFIR-2026-0618**. Security Operations Center (SOC) telemetry validated that an isolated host system (**192.168.92.133**) interacted with a malicious tracking URL dispatched via a GoPhish orchestration framework on **192.168.56.101**. Forensic artifact extraction via Wireshark PCAP payloads proved cleartext credential transmission over port 8080. Total host containment and network isolation were achieved in under 4 hours, ensuring zero data leakage or lateral movement. To transition the perimeter from reactive alerting to proactive interception, a custom-engineered multi-layered Sequential Artificial Neural Network (ANN) deep learning model was deployed to dynamically block subsequent delivery phases.

3. INCIDENT LIFECYCLE & TIMELINE (SANS BLUEPRINT)

- Preparation:** Establishment of advanced telemetry configurations, logging baselines, Sysmon system rules, and centralized Splunk SIEM ingestion schemas prior to execution.
- Identification:** Anomalous outbound web connections triggered a high-severity alert. Analysis of network PCAP data revealed explicit HTTP POST requests delivering plain text credentials with the

tracker token ?rid=3D17FCWja.

3. **Containment:** Executed swift host-level isolation from the main laboratory gateway. Deployed strict endpoint firewall blocks to entirely drop outbound communication directed toward network node **192.168.56.101**.
4. **Eradication:** Completely purged local web application caches, session cookies, and persistent browser memory states. Audited local registry hives and Scheduled Tasks to guarantee zero active adversarial hooks.
5. **Recovery:** Validated operating system integrity matrices, re-established secure monitored gateway routing, and initialized an intensive 48-hour network surveillance state.
6. **Lessons Learned:** Conducted a post-incident review to catalog detection deficiencies, resulting in the implementation of an automated deep-learning validation pipeline for incoming URL structures.

4. DIGITAL EVIDENCE INVENTORY & PRESERVATIONS

- **Artifact E-01 (Network Capture):** Wireshark PCAP files capturing raw packet sequences showing the unencrypted transmission of credential input directly to the external server.
- **Artifact E-02 (Endpoint Security Logs):** Windows Microsoft-Windows-Sysmon/Operational logs confirming **Event ID 3 (Network Connection)**, linking browser actions directly to the hostile external port.
- **Artifact E-03 (SIEM Dashboard Analytics):** Aggregated Splunk search metrics matching the exact network connection timestamps to the GoPhish server ingress activity records.

5. PROACTIVE AI CONTROL DEPLOYMENT

Because traditional signature-based perimeters struggle to counter highly dynamic look-alike tracking URLs, an advanced defensive measure was implemented. Classic linear machine learning models exhibited sub-optimal accuracy boundaries when parsing high-dimensional URL string sequences. To resolve this, a specialized **Sequential Artificial Neural Network (ANN)** deep learning model was built to map non-linear contextual indicators. This model was integrated into a production-ready Graphical User Interface (GUI), providing incident handling teams with an instantaneous, automated safety verdict classification mechanism.

6. STRATEGIC RECOMMENDATIONS & ACTIONS

- Embed the optimized deep learning Sequential ANN infrastructure directly into the edge secure email gateway processing filters.
- Migrate high-risk operational profiles to phishing-resistant cryptographic authentication methods (FIDO2/WebAuthn hardware keys).

- Configure automated playbook orchestration triggers within Splunk to immediately enforce host network isolation upon any validated Sysmon Event ID 3 alert directed toward unverified subnet spans.

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